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By Amit Katiyar (MCA-JNU)

MATHEMATICS

1. A professor has 24 text books on computer science and is concerned about their coverage of the topics (P) compilers, (Q) data structures and (R) operating systems. The following data gives the number of books that contain material on these topics : n(P)= 8, n(Q)=13, $n(R) = 13, n(P \cap Q) = 3, n(P \cap Q \cap R) = 2$, where n(x) is the cardinality of the set x, then, the number of text books that have no material no compiler is
(a)4 (b)8 (c)12 (d)None

2. The value of $tan\left(\frac{7\pi}{8}\right)$ is
(a)$1-\sqrt{2}$ (b)$1+\sqrt{2}$
(c)$\sqrt{2}+\sqrt{3}$ (d)$\sqrt{2}-\sqrt{3}$

3. If **a** and **b** are vectors such that $|a| = 13, |b| = 5$ and $a.b = 60$, then the value of $|a \times b|$ is
(a)625 (b)225 (c)45 (d)25

4. Two towers face each other separated by a distance of 25m. As seen from the top of the first tower. The angle of depression of the second tower's base is 60° and that of the top is 30°. The height (in meters) of the second tower is
(a)$\frac{50}{\sqrt{3}}$ (b)$\frac{25}{\sqrt{3}}$ (c)50 (d)$25\sqrt{3}$

5. If $a = 4\hat{\imath} + 6\hat{\jmath}$ and $b = 3\hat{\imath} + 4\hat{k}$,then the vector form of the component of a along b is
(a)$\frac{18}{10\sqrt{3}}(3\hat{\imath}+4\hat{k})$ (b)$\frac{18}{5}(3\hat{\imath}+4\hat{k})$
(c)$\frac{18}{\sqrt{3}}(3\hat{\imath}+4\hat{k})$ (d)$\frac{12}{25}(3\hat{\imath}+4\hat{k})$

6. The value of $\sin^{-1}\frac{1}{\sqrt{2}} + \sin^{-1}\frac{\sqrt{2}-\sqrt{1}}{\sqrt{6}} + \sin^{-1}\frac{\sqrt{3}-\sqrt{2}}{\sqrt{12}} + \cdots\ldots$ to infinity is equal to
(a)π (b)$\frac{\pi}{3}$ (c)$\frac{\pi}{2}$ (d)$\frac{\pi}{4}$

7. If two circle $x^2+y^2+2gx+2fy=0$ and $x^2+y^2+2g'x+2f'y=0$ touch each other, the which of the following is true?
(a)gf = gf′ (b)g′f = gf′
(c)gg′ = ff′ (d)None of these

8. $\int_0^x [\cot x]\, dx$, where $[\circ]$ denotes the greatest integer function, is equal to
(a)$\frac{\pi}{2}$ (b)1 (c)-1 (d)$-\frac{\pi}{2}$

9. In a right- angled triangle, the hypotenuse is four times the perpendicular drawn to it form the opposite vertex. The value of one of the acute angles is
(a)45° (b)30° (c)15° (d)None

10. A a targeting B, B and C are targeting A. Probability of hitting the target by A, B and C are $\frac{2}{3},\frac{1}{2}$ and $\frac{1}{3}$ respectively. If A is hit, then the probability that B hit the target and C does not, is
(a)$\frac{1}{2}$ (b)$\frac{1}{3}$ (c)$\frac{2}{3}$ (d)$\frac{3}{4}$

11. If the angles of a triangle are in the ratio 2:3:7, then the ratio of the sides opposite to these angles is
(a)$\sqrt{2}: 2: \sqrt{3}+1$ (b)$2: \sqrt{2}: \sqrt{3}+1$
(c)$2: \sqrt{2}: \frac{\sqrt{2}}{\sqrt{3}-1}$ (d)$\frac{1}{\sqrt{2}}: 2: \frac{\sqrt{3}+1}{2}$

12. Suppose that, A and B are two events with probabilities$P(A) = \frac{1}{2}, P(B) = \frac{1}{3}$, then, which of the following is true?
(a) $\frac{1}{3} \le P(A\cap B) \le \frac{1}{2}$ (b) $\frac{1}{4} \le P(A\cap B) \le \frac{1}{3}$
(c) $\frac{1}{6} \le P(A\cap B) \le \frac{1}{3}$ (d) $\frac{1}{4} \le P(A\cap B) \le \frac{1}{2}$

13. The number of one –to- one function form$\{1,2,3\}$ to $\{1,2,3,4,5\}$ is
(a)125 (b)243
(c)10 (d)60

14. A harbour lies in a direction 60° South of West from a fort and at a distance 30 km from it, a ship sets out from the harbour at noon and sails due East at 10 km an hour. The time at which the ship will be 70 km from the fort is
(a)7 PM (b)8PM
(c)5PM (d)10PM

15. If x, y, and z are three consecutive positive integers, then $\log(1+xz)$ is
(a)$\log y$ (b)$\log\frac{y}{2}$
(c)$\log(2y)$ (d)$2\log(y)$

16. If a, b and c are in geometric progression, then $\log_{ax} x, \log_{bx} x$ and $\log_{cx} x$ are in
(a)arithmetic progression
(b)geometric progression
(c)harmonic progression
(d)arithmetico-geometric progression

17. If a and b are vector in space, given by $a = \frac{\hat{\imath}-2\hat{\jmath}}{\sqrt{5}}$ and $b = \frac{2\hat{\imath}+\hat{\jmath}+3\hat{k}}{\sqrt{14}}$, then the value of$(2a+b).[(a\times b)\times(a-2b)]is$
(a)3 (b)4
(c)5 (d)6

18. The value of the sum$\frac{1}{2\sqrt{1}+1\sqrt{2}}+\frac{1}{3\sqrt{2}+2\sqrt{3}}+\frac{1}{4\sqrt{3}+3\sqrt{4}}+\cdots+\frac{1}{25\sqrt{24}+24\sqrt{25}} is$
(a)$\frac{9}{10}$ (b)$\frac{4}{5}$
(c)$\frac{14}{15}$ (d)$\frac{7}{15}$

19. If $a = \hat{\imath}-\hat{k}, b = x\hat{\imath}+\hat{\jmath}+(1-x)\hat{k}$ and $c = y\hat{\imath}+x\hat{\jmath}+(1+x-y)\hat{k}$, then [a b c] depends on
(a)neither x nor y (b)Only x
(c)Only y (d)Both x and y

20. If $42({}^nP_2) = {}^nP_4$, then the value of n is
(a)2 (b)4
(c)9 (d)42

21. Let a and b be two vectors. Which of the following vectors are not perpendicular to each other?
(a)$(a\times b)$ and a (b)$(a+b)$ and $a\times b$
(c)$(a+b)$ and $a-b$ (d)$a-b$ and $a\times b$

22. If $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$,where a, b and c are real positive numbers such that $abc = 1$ and $A^TA = 1$, then the equation that holds true among the following is
(a) $a+b+c=1$ (b) $a^2+b^2+c^2=1$
(c) ab + bc + ca = 0(d) $a^3+b^3+c^3=4$

23. The equation of the tangent at any point of curve $x = a\cos 2t, y = 2\sqrt{2}a\sin t, with\ m$ as its slope is
(a) $y = mx + a\left(m - \frac{1}{m}\right)$
(b) $y = mx - a\left(m + \frac{1}{m}\right)$
(c) $y = mx + a\left(a + \frac{1}{a}\right)$
(d) $y = amx + a\left(m - \frac{1}{m}\right)$

24. The locus of the mid-point of all chords of the parabola $y^2 = 4x$, which are drawn through its vertex is
(a)$y^2 = 8x$ (b)$y^2 = 2x$
(c)$x^2 + 4y^2 = 16$ (d)$x^2 = 2y$

25. The value of $\lim_{x\to a}\frac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3a-x}-2\sqrt{x}}$ is
(a)$\frac{2}{3}$ (b)$\frac{2}{\sqrt{3}}$
(c)$\frac{3\sqrt{3}}{2}$ (d)$\frac{2}{3\sqrt{3}}$

26. The value of $\int_{-\pi/3}^{\pi/3}\frac{x\sin x}{\cos^2 x}dx$ is
(a)$\frac{1}{3}(4\pi + 1)$ (b)$\frac{4\pi}{3} - 2\log tan\frac{5\pi}{12}$
(c)$\frac{4\pi}{3} + \log tan\frac{5\pi}{12}$ (d)$\frac{4\pi}{3} - \log tan\frac{5\pi}{12}$

27. The foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1\ and\ the\ hyperbola\ \frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide, then value of b^2 is
(a)1 (b)5
(c)7 (d)9

28. If $A + B + C = \pi$, Then the value of $\begin{vmatrix}\sin(A+B+C) & \sin B & \cos C\\ -\sin B & 0 & \tan A\\ \cos(A+B) & -\tan A & 0\end{vmatrix}$ is
(a)0 (b)1
(c)$2\sin A sin B$ (d)2

29. If the mean deviation of the number $1, 1 + d, 1 + 2d \ldots 1 + 100d$ from their mean is 225, then the value of d is
(a)20.0 (b)10.1
(c)20.2 (d)8.91

30. If $P = \sin^{20}\theta + \cos^{48}\theta$, then the inequality that holds for all values of θ is
(a)$P \geq 1$ (b)$0 < P \leq 1$
(c)$1 < P < 3$ (d)$0 \leq P \leq 1$

31.The foot of the perpendicular from the point(2,4) upon $x + y = 1$ is
(a)$\left(\frac{1}{2}, \frac{3}{2}\right)$ (b)$\left(-\frac{1}{2}, \frac{3}{2}\right)$
(c)$\left(\frac{4}{3}, \frac{1}{2}\right)$ (d)$\left(\frac{4}{3}, -\frac{1}{2}\right)$

32. The value of k for which the equation $(k-2)x^2 + 8x + k + 4 = 0$ has both real, distinct and negative and negative roots is
(a)0 (b)2
(c)3 (d)-4

33. If $(2,1), (-1,-2), (3,3)$ are the mid-points of the sides BC, CA and AB of a triangle ABC, then equation of the line BC is
(a)$5x + 4y + 6 = 0$ (b)$5x - 4y - 6 = 0$
(c)$5x + 4y - 6 = 0$ (d)$5x - 4y + 6 = 0$

34. If a fair dice is rolled successively, then the probability that 1 appears in an even numbered throw is
(a)$\frac{5}{36}$ (b)$\frac{3}{11}$
(c)$\frac{1}{6}$ (d)$\frac{5}{11}$

35. Let $a = \hat{\imath} + \hat{\jmath} + \hat{k}, b = \hat{\imath} - \hat{\jmath} + \hat{k}\ and\ c = \hat{\imath} - \hat{\jmath} - \hat{k}$ be three vectors. A vector v in the plane of **a** and **b** whose projection on $\frac{c}{|c|}\ is\ \frac{1}{\sqrt{3}}, is$
(a)$3\hat{\imath} - \hat{\jmath} + 3\hat{k}$ (b)$\hat{\imath} - 3\hat{\jmath} + 3\hat{k}$
(c)$5\hat{\imath} - 2\hat{\jmath} + 5\hat{k}$ (d)$2\hat{\imath} - \hat{\jmath} + 3\hat{k}$

36.The number of bit strings of length 10 that contain either five consecutive 0's or five consecutive 1's is
(a)64 (b)112
(c)220 (d)222

37. $If 0 < x < \pi\ and \cos x + \sin x = \frac{1}{2}, then\ the\ value\ of \tan x$
(a)$\frac{4-\sqrt{7}}{3}$ (b) $\frac{-4+\sqrt{7}}{3}$
(c)$\frac{1+\sqrt{7}}{4}$ (d)$\frac{1-\sqrt{7}}{4}$

38. If a, b and c are the position vectors of the vertices A, B, C of a triangle ABC, then the area of the triangle ABC is
(a) $\frac{1}{2}|a \times b + b \times c + c \times a|$
(b) $|a \times b|$
(c) $\frac{1}{2}|a \times b - b \times c - c \times a|$
(d) $|a \times (b \times c)|$

39. If $\int e^x(f(x) - f'(x))dx = \phi(x)$, is $then\ the\ value\ of \int e^x f(x)dx$
(a) $\phi(x) + e^x f(x)$ (b) $\phi(x) - e^x f(x)$
(c)$\frac{1}{2}[\phi(x) + e^x f(x)]$ (d)$\frac{1}{2}[\phi(x) + e^x f'(x)]$

40. If $3x + 4y + k = 0$ is tangent to the hyperbola $9x^2 - 16y^2 = 144$, then value of K is
(a)0 (b)1 (c)-1 (d)-3

41. a, b, and c are positive integers such that $a^2 + b^2 - 2bc = 100\ and\ 2ab - c^2 = 100$. Then, the value of $\frac{a+b}{c}$ is
(a)10 (b)100 (c)2 (d)20

42. If $(-4,5)$ is one vertex and $7x - y + 8 = 0$ is one diagonal of a square, then the equation of the other diagonal is
(a)$x + 7y = 21$ (b)$x + 7y = 31$
(c)$x + 7y = 31$ (d)$x + 7y = 35$

43. Out of $2n + 1$ ticket, which are consecutively numbered, three are drawn at random. Then, the probability that, the numbers on them are in arithmetic progression is
(a)$\frac{n^2}{4n^2-1}$ (b)$\frac{n}{4n^2-1}$
(c)$\frac{3n^2}{4n^2-1}$ (d)$\frac{3n}{4n^2-1}$

44. A circle touches the X- axis and also touches another circle with centre at(0,3) and radius 2. Then the locus of the centre of the first circle is
(a)a parabola (b)a hyperbola
(c)a circle (d)an ellipse

45. Let $\bar{P}$ and $\bar{Q}$ denote the complements of two sets P and Q. Then, the set $(P-Q)\cup(Q-P)\cup(P\cap Q)$ is
(a)$P\cup Q$ (b)$\bar{P}\cup\bar{Q}$s
(c)$P\cap Q$ (d)$\bar{P}\cap\bar{Q}$

46. With the usual notation, $\frac{d^2x}{dy^2}$ is
(a)$\left(\frac{d^2y}{dx^2}\right)^{-1}$ (b)$\frac{d^2y}{dx^2}\left(\frac{dy}{dx}\right)^{-2}$
(c)$-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-2}$ (d)$-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$

47. The radius of the circle passing through the foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ and having its centre at (0,3)is
(a)4 units (b)3 units
(c)$\sqrt{12}units$ (d)$\frac{7}{2}$ units

48. A function $F:(0,\pi)\to R$ defined by $f(x)=2\sin x+cos2x$ has
(a) A local minimum but no local maximum
(b) A local maximum but no local minimum
(c)Both local minimum and local maximum
(d)Neither a local minimum nor a local maximum

49. A matrix M_r is defined as $M_r=\begin{bmatrix} r & r-1 \\ r-1 & r\end{bmatrix} r\in N$, then the value of $\det(M_1)+\det(M_2)+\cdots+\det(M_{2015})$ is
(a)2014^2 (b)2013^2
(c)2015 (d)2015^2

50. If$AC=2\hat{\imath}+\hat{\jmath}+\hat{k}$ and $BD=-\hat{\imath}+3\hat{\jmath}+2\hat{k}$, then area of the quadrilateral ABCD is
(a)$\frac{5}{2}\sqrt{3}$ (b)$5\sqrt{3}$
(c)$\frac{15}{2}\sqrt{3}$ (d) $10\sqrt{3}$

LOGICAL REASONING

Directions **[Q. Nos. 51-54]** A circular field with inner radius of 10 m and outer radius of 20 m is divided into 5 successive stage for ploughing. The ploughing at each stage, with starting points P1, P2, P3, P4 and P5 was is allotted to one of the five farmers F1,F2,F3,F4 and F5 not necessarily in that order.

- F5 was alloted the stage starting a point P4.
- The stage from P5 to P3 was not the first stage.
- F4 was allotted the work of the fourth stage.
- Finishing point of stage 3 was P1 and the work was not allotted to F1.
- F3 was allotted the work of stage ending at P5.

51. Which of the following is the finish point for farmer F2?
(a)P1 (b)P2
(c)P3 (d)P4

52. Which stage was ploughed by F5
(a)2 (b)3
(c)4 (d)5

53. What are the starting and ending points of the field ploughed by F4?
(a)P1 and P2 (b)P1 and P4
(c)P4 and P2 (d)P2 and P4

54. What is the starting point for stage 3?
(a)P2 (b)P3
(c)P4 (d)Cannot be determined

55. If Tuesday falls on the fourth of a month, then which day will fall three days after 24th of the same month?
(a)Monday (b)Tuesday
(c)Thursday (d)Friday

56. If the statements "All chickens are birds", "Some chickens are hens" and "Female birds lay eggs", are all facts, then which of the following must also be a fact?
I. All birds lay eggs
II. Some hens are birds
III. Some chickens are not hens
(a)I and II (b)II and III
(c)I and III (d)Neither I nor II nor III

Directions [Q. Nos. 57-60]

- There is a family of six members A, B, C, D, E and F.
- There are two married couples in the family and the family members represent three generations.
- Each member has a distinct choice of a colour amongst Green, Yellow, Black, Red, White and Pink.
- No lady members likes either Green or White.
- C, who likes Black colour, is the daughter-in-law of E.
- B is the brother of F and son of D and likes Pink.
- A is the grandmother of F and F does not like Red.
- Wife of the husband having a choice for Green colour likes Yellow.

57. Which of the following is the colour preference of A?
(a)Red (b)Yellow
(c)Either Red or Yellow (d)Cannot be determined

58. Which of the following could be the colour combination of one of the couples?
(a)Yellow-Red (b)Green-Black
(c)Red-Yellow (d)Yellow-Green

59. Which of the following is one of the married couples?
(a)CD (b)AC
(c)AD (d)Cannot be determined

60. Which of the following is true about F?
(a)Brother of B (b)Sister of B
(c)Daughter of C (d)Cannot be determined

61. If the English word "EXAMINATION" is coded as 56149512965, then the word "GOVERNMENT" is coded as
(a)7645954552 (b)7654694562
(c)7645955423 (d)7654964526

62. Gopal starts from his house towards West. After walking a distance 30m, he turned towards right and walked 20m. He turned left and after moving a distance of 10m, turned to his left again and walked 40m. Then, he turned left and walked 5m. Finally, he turns to his left. In which direction is the walking now?
(a)North (b)South
(c)East (d)South East

63. Read the conclusion and then decide which of the given conclusions logically follows from the two given statements (i)and (ii)disregarding commonly known facts.

Statements:
(i) No women teacher can play.
(ii) Some women teachers are athletes.
Conclusions:
(i)Male athletes can play.
(ii) Some athletes can play.
(a)Only conclusion I follows
(b)Only conclusion II follows
(c)Either I or II follows
(d)Neither I nor II follows

64. Which of the following come next in the series 8, 6, 9, 23, 87,.....?
(a)28 (b)226 (c)324 (d)429

65. In an examination there are 100 questions divided into 3 parts A, B and C and each part should contain atleast one question. Each question in parts A, B and C carry 1, 2 and 3 marks respectively. Part A is for atleast 60% of the total marks and part B should contain 23 questions. How many questions must be set in part C?
(a)1 (b)2
(c)3 (d)Cannot be determined

66. If ÷ means addiction, - means division, × means subtraction and + means multiplication, then the value of $\frac{(36\times4)-8\times4}{4+8\times2+16\div1}$ is
(a)0 (b)8
(c)12 (d)16

67. Which letter is the word CYBERNETICS occupies the same position as it does in the English alphabet?
(a)C (b)E
(c)I (d)T

68. The remainder when 2^{31} is divided by 5 is
(a)1 (b)2
(c)3 (d)4

Directions[Q. Nos. 69-70] the letter of English alphabet from A to M were written, leaving space for one letter between every two letters and then the remaining letters were inserted beginning with N and ending the series with Z after M.

69. Which letter would be 3rd to the right of the 7th letter from the left?
(a) C (b) O
(c) R (d)S

70. Which letter would be exactly in the middle of eighteenth letter from the beginning and fifteenth from the end?
(a)G (b)H
(c)J (d)L

71. How many 3- digit numbers divisible by 5, can be formed using the digits 2 ,3 ,5 ,6 ,7 and 9, without repetition of digits?
(a)216 (b)20
(c)17 (d)18

72. Using only 2, 5, 10, 25 and 50 paise coins, what is the smallest number of coins required to pay exactly 78paise, 69 paise and Rs.1.01 to three different persons?
(a)19 (b)20
(c)17 (d)18

73. Which of the following two patterns will fit in the blanks of the series $ZA_5, Y_4B, XC_6, W_3D, \ldots\ldots?$
(a)$\mathrm{VE_7}$ and$\mathrm{U_2E}$ (b)$\mathrm{V_2E}$ and $\mathrm{U_7F}$
(c)$\mathrm{VE_7}$ and $\mathrm{U_2F}$ (d)$\mathrm{VF_7}$ and $\mathrm{U_2E}$

74. Which of the following numbers comes next in the two-digit decimal number sequence 61,52,63,94,.......?
(a)65 (b)64
(c)56 (d)46

75. Three ladies X, Y and Z marry three man A, B and C, X is married to A, Y is not married to an engineer. Z is not married to a doctor, C is not a doctor and A is a lawyer. Then, which of the following statements is correct?
(a) X is married to a doctor
(b) Y is married to C, who is a doctor
(c) Z is married to B, who is an engineer
(d) None of the above

76. There are five books A, B, C, D and E placed on a table .If A is placed below E, C is placed D, B is placed below A and D is placd between A and E, then which of the following books can be on the top?
(a) D or E (b) C or E
(c)A or E (d)None of these

77. Among five children A, B, C, D and E, B is taller than E but shorter than D. A is shorter than C but taller than D. If all the children stand in a line according to their heights, then who would be the fourth if counted from the tallest one?
(a)D (b)C
(c)B (d)A

Directions[Q.Nos.78-81]
- In a family of six person A, B, C, D, E and F there are two married couples.
- D is grandfather of A and mother B.
- C is wife of B and mother F.
- F is the granddaughter of E.

78. What is C to A?
(a) Daughter (b)Grandmother
(c) Mother (d) Cannot be determined

79. How many male members are there in the family?
(a) Two (b) Three
(c) Four (d) Cannot be determined

80. Who among the following in one of the couples?
(a)CD (b)DE
(c)EB (d)Cannot be determined

81. Which of the following is true?
(a)A is brother of F (b)A is sister of F
(c)B has two daughters (d)None of these

Directions [Q. Nos. 82-85] A, B, C,D, E, F and G are seven girls having different amount of money from among Rs.10,20,40,60,80,120 and 200 with them . They had 3 chocolates, 2 toffees and 2 lollipops together, each other having one of these seven items.

B and F do not have chocolates and they have Rs.200 and Rs.80, respectively.

- C has Rs.60 with her and G has an amount which neither Rs.40 nor Rs.120.
- A has Rs10 and does not have a toffee.
- The girl having Rs.40 with her is the only one other than A to have the same type of item.
- E and girl having Rs.20 with her have the same find of item.

82. How much amount does G have with her?
(a) Rs.20 (b) Rs.10
(c) Rs.60 (d) None of these

83. Which of the following girl have chocolates with items?
(a) F, C, G (b) C, G, E
(c) C, G, D (d) G, D, E

84. Which of the following combination is definitely correct?
(a)C-chocolate Rs.60 (b)G- toffee-Rs.20
(c) C-chocolate Rs.40 (d)None of these

85. Which girls has Rs.40 with her?
(a)E (b)A
(c)D (d)None of these

86. P,Q,R,S,T,U and V are sitting in a row facing North. In order determine, who is sitting exactly in the middle of the row, which of the following information is needed?
I. T and U are sitting at extreme ends of the row.
II. S is third to the right of T
III. Q is four places to the left of R and P is two places of the left of V
(a) I and II only are sufficient
(b) I and III only are sufficient
(c) I and either II or III are sufficient
(d) I, II and III

87. How many times do the hour and the minute hands of a clock overlap in 24 h?
(a) 24 (b) 22
(c) 26 (d) 20

88. In a certain code, TOGETHER is coded as RQEGRJCT. In the same code, PAROLE will be written as
(a) NCPQJG (b) NCQPJG
(c) RCPQJK (d) RCTQNG

89. A drawer contains 10 black and 10 brown socks which are all mixed up. What is the smallest number of socks to be taken from the drawer to decided without seeing them, to be sure that there is atleast one pair of socks of the same colour ?
(a)11 (b)10
(c)3 (d)Cannot be determined

90. Find the missing number in the series: 4, 7, 25, 10, ... , 20, 16, 19
(a)13 (b)15
(c)20 (d)28

GENERAL ENGLISH

91. Identify the type of error in the sentence: "The cost of this project will be much lesser than 5% more than that predicted earlier".
(a) syntactical error (b) punctuation error
(c) grammatical error (d) conflicting words

92. Insert appropriate prepositions in the blanks to complete the sentence. " This property has been ...... the possession.......the royal family.....generations."
(a) with, of, of (b) in, of, for
(c) in, with, by (d) of, by, since

93. Choose the right word to fill in the blank in the sentence. "The mermaid legend....... Have originated with a group of mammals collectively known to science as Srinians".
(a)should (b)may
(c)need (d)can

94. Identify appropriate word to fill the blanks in the sentence. "The feeling of guilt left a ...... impression in the life?"
(a)perennial (b)parennial
(c)perannial (d)perinial

95. Which of the following sentences is grammatically incorrect?
(a)He is smiling (b)He smiles
(c)He always smiles (d)He is always smiling

96. Find the most suitable phrasal verb to be filled in the following sentence.
"Left too long in the sun, the leaves had all"
(a) Shrugged off (b) Shared out
(c) Shrivelled up (d) Skived off

97. Fill in the blank from among the choices in the sentence.
A 'Couch potato' is a person who.......
(a) spends a lot of time watching television
(b) spends money on potatoes
(c) likes potatoes
(d) is lazy, but intelligent

98. Which of the following sentence is grammatically incorrect?
(a) She never travelled abroad for fear of becoming ill through eating foreign food
(b) She avoids foreign travel as she fears she will become ill through eating foreign food
(c) She never travelled abroad due to her fear of becoming ill through eating foreign food
(d) She never travelled abroad in fear for becoming ill with eating foreign food

99. Match the most suitable phrasal Verb from Group L to each word in Group M.

Group L	Group M
1. Call out	P. A Footballer
2. Stands in for	Q. A Criminal
3. Send down	R. A Colleague
4. Send off	S. A Doctor

(a) 3-R, 2-S, 1-P, 4-Q (b)1-S, 2-R, 3-Q, 4-P
(c)1-P, 2-Q, 3-R, 4-S (d)2-P, 3-S,4-R, 1-Q

100. Choose the one which best expresses the following sentence in passive/active voice.
"You can play with these kittens quite safely"
(a) These kittens can be played with quite safely
(b) These kittens can play with you quite safely
(c) These kittens can be played with you quite safely
(d) These kittens can played with quite safely

101. Which of the following terms refers to the original inhabitants of a place?
(a) Originals (b) Aborigines
(c) Abominablesss (d) Cannibals

102. Replace the underlined word with one of the choice given without changing the meaning of the sentence. "The news of our success was met with exuberant cries"
(a) Excited (b) Pathetic
(c) Exclusive (d) Poignant

103. Select the word that is furthest is meaning to the word AFFLUENCE.
(a) Stagnation (b) Misery
(c) Neglect (d) Poverty

Directions [Q. Nos. 104-106]: The proud warrior class of the samurai (meaning "those who serve") grew from a band of mercenaries hired by feudal landowners in the 11th century to win them the control of Honshu, Japan's main island. These mercenaries lived by the cult of the sword, worshipping athletic prowess and martial skills. They developed a fierce loyalty to their masters and a fearlessness that made them formidable adversaries. They fought in elaborate armour, wielding their most prized possession, a double-edged sabre with which they could cut a man in half.

Later the spartan principles of Zen Buddhism, with its love of nature softened their fighting zeal. It became fashionable for them to live sparce and frugal lives during the Kamakura era (1192-1333), when the ruling warrior family Minamato moved their seat to power of the eastern city of Kamakura.

104. Who are usually referred to as mercenaries?
(a) Soldiers with martial skills
(b) proud warriors
(c) soldiers who fight for money
(d) Loyal warriors

105. Which of the following best describes the warriors?
(a) proud, greedy (b) Fearless, worshipful
(c) Loyal, fearless (d)possessive, soft

106. In the kamakura period it became fashionable for these warriors to live
(a) Zealour lives (b) Austere lives
(c) Powerful lives (d) Natural lives

107. Rearrange the parts of a sentence referred to by P, Q, R and S to form a complete and meaningful sentence
"I enclose ___ _ ___ ___ ____"
P: and the postage
Q: a postal order
R: the price of books
S: which will cover
(a) RPSQ (b) QSPR
(c) QSRP (d) QPSR

108. Which of the following is the antonym of the word 'Exigency'?
(a) Penchant (b) Emergency
(c) Earnestness (d) Indifference

109. Which of the following prepositions fills up the blanks in the sentence?
"Quinine is an effective antidote Malaria".
(a) to (b) against
(c) for (d) None of these

110. In the sentence, "The defence labs have showcased many new innovations this year......there is an error of
(a) redundancy (b) word order
(c) collocation (d) omission

COMPUTER AWARENESS

111. When the value 37H is divided by 17H , the remainder is
(a) COH (b) 03H
(c) 07H (d) 09H

112. The number of boolean function possible with n binary variables is equal to
(a)2^{2^n} (b)2^n
(c)$2^{2^{n+1}}$ (d)2^{n+1}

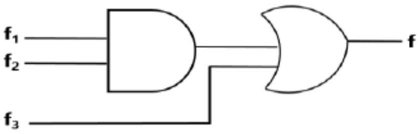
113. Consider 4-bit gray code representation of a numbers. Let $h_3h_2h_1h_0$ be the gray representation of a number n and $g_3g_2g_1g_0$ be the gray code representation of the number($n+1$) modulo 15. Which of the following functions is correct?
(a) $g_0(h_3h_2h_1h_0) = \sum(1,2,3,6,10,13,14,15)$
(b) $g_1(h_3h_2h_1h_0) = \sum(4,9,10,11,12,13,14,15)$
(c) $g_2(h_3h_2h_1h_0) = \sum(2,4,5,6,7,12,13,15)$
(d) $g_3(h_3h_2h_1h_0) = \sum(0,1,6,7,10,11,12,13)$

114. The minimum number NAND gates required to realize $AB + AB'C + AB'C'$ is
(a)3 (b)2
(c)1 (d) 0

115. Which optical phenomenon is utilized in the operation of the latest write- once optical storage medium called digital paper?
(a)Polarisation (b)Interference
(c)Internal reflection (d)Diffraction

116. P is a 16- bit signed integer. The 2's complement representation of P is $(F87B)_{18}$. The 2's complement representation of 8P is
(a)$(\text{C3D8})_{16}$ (b)$(187B)_{16}$
(c)$(188B)_{16}$ (d)$(987B)_{16}$

117. Given,f_1, f_3 and f in canonical sum of products form (in decimal) from the circuit



$f_1 = \sum m(4,5,6,7,8), f_3 = \sum m(1,6,15) and\ f = \sum m(1,6,8,15)$, then f_2 is
(a) $\sum(4,6)$ (b) $\sum$ (4,8)
(c) $\sum(6,8)$ (d) $\sum(4,6,8)$

118. Which of the following is equivalent to the expression $(\overline{X+Y+Z})$?
(a)$(\bar{X}+\bar{Y})Z$
(b)$(X+Y)\bar{Z}$
(c)$(\bar{X}+\bar{Y})\bar{Z}$
(d)$(X+Y)Z$

119. $\{p \to q \vee r, q \to s, r \to s\}$ is logically equivalent to
(a)$q \to r$ (b)$r \to q$
(c)$p \to s$ (d)$s \to p$

120. The minimum number of MOS transistors required to make a dynamic RAM cell is
(a)1 (b)2
(c)3 (d)4

ANSWER KEY

1. (d)	21. (c)	41. (c)	61. (a)	81. (d)	101. (b)
2. (a)	22. (b)	42. (b)	62. (b)	82. (a)	102. (a)
3. (d)	23. (b)	43. (d)	63. (d)	83. (b)	103. (b)
4. (a)	24. (b)	44. (a)	64. (d)	84. (a)	104. (a)
5. (d)	25. (d)	45. (a)	65. (a)	85. (c)	105. (c)
6. (c)	26. (b)	46. (d)	66. (a)	86. (a)	106. (b)
7. (c)	27. (c)	47. (a)	67. (c)	87. (b)	107. (c)
8. (c)	28. (a)	48. (c)	68. (c)	88. (a)	108. (b)
9. (c)	29. (d)	49. (d)	69. (c)	89. (c)	109. (a)
10. (a)	30. (a)	50. (b)	70. (b)	90. (a)	110. (d)
11. (a)	31. (b)	51. (a)	71. (b)	91. (a)	111. (d)
12. (b)	32. (c)	52. (d)	72. (a)	92. (b)	112. (b)
13. (d)	33. (b)	53. (b)	73. (c)	93. (b)	113. (b)
14. (b)	34. (d)	54. (b)	74. (d)	94. (a)	114. (b)
15. (d)	35. (a)	55. (c)	75. (d)	95. (d)	115. (d)
16. (a)	36. (d)	56. (b)	76. (d)	96. (c)	116. (a)
17. (c)	37. (b)	57. (b)	77. (c)	97. (a)	117. (c)
18. (b)	38. (a)	58. (d)	78. (c)	98. (d)	118. (d)
19. (a)	39. (c)	59. (a)	79. (d)	99. (b)	119. (c)
20. (c)	40. (a)	60. (a)	80. (d)	100. (b)	120. (a)

SOLUTION

MATHS

1. **(d)** No materials of compiler.

= Total books – Number of books with compiler.

= 24 – 8 = 16.

2. **(a)**

$\tan\left(\frac{7\pi}{8}\right) = \tan\left(\pi - \frac{\pi}{8}\right) = -\tan\frac{\pi}{8}$

We know that

$\tan\frac{\pi}{8} = \sqrt{2} - 1 \Rightarrow -\tan\frac{\pi}{8} = -(\sqrt{2}-1) = 1 - \sqrt{2}$

3. (d)

|a| = 13, |b| =5, a.b = 60

|a x b| = $\sqrt{|a|^2.|b|^2 - (a.b)^2}$

|a x b| =$\sqrt{13^2 \times 5^2 - (60)^2}$

= $\sqrt{169 \times 25 - 3600}$

= $\sqrt{4425 - 3600}$

= $\sqrt{625}$ = 25

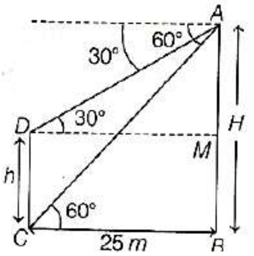
4. **(a)** $AB = H$ (say)

$\Rightarrow \tan 30^\circ = \frac{AM}{25} \Rightarrow AM = \frac{25}{\sqrt{3}}$

$\Rightarrow \tan 60^\circ = \frac{AB}{25}$

$\Rightarrow \sqrt{3} = \frac{AM+MB}{25} = \frac{25/\sqrt{3}+MB}{25}$

$\Rightarrow MB = \frac{50}{\sqrt{3}}$



$\Rightarrow\ AB = H = AM + MB = \frac{25}{\sqrt{3}} + 25\left(\frac{2}{\sqrt{3}}\right)$

$H = 25\sqrt{3}$

$\Rightarrow$ Height of 2nd tower $= h = CD = MB = \frac{50}{\sqrt{3}}$

5. **(d)**

Since a = 4i + 6j and b = 3i + 4k

Vector component of a along b is $\frac{(a.b)b}{b}$

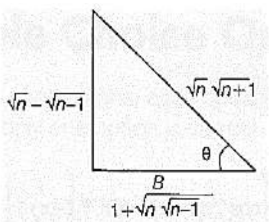
$= \frac{(4i+6j).(3i+4k)}{\left(\sqrt{3^2+4^2}\right)^2}\ (3i + 4R)$

$\Rightarrow \frac{12}{25}\ (3i + 4k)$

6. **(c)**

$\sin^{-1}\frac{1}{\sqrt{2}} + \sin^{-1}\frac{\sqrt{2}-\sqrt{1}}{\sqrt{6}} + \sin^{-1}\frac{\sqrt{3}-\sqrt{2}}{\sqrt{12}} + \cdots$

$T_n = \sin^{-1}\frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n}\sqrt{n+1}} = \theta \Rightarrow \frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n}\sqrt{n+1}}$ Windows $= \sin\theta$



Base $^2 = n(n+1) - (\sqrt{n} - \sqrt{n-1})^2$

$= n(n+1) - (n + \square - 1 - 2\sqrt{n(n-1)})$

$= n^2 + n - 2n + 1 + 2\sqrt{n(n-1)}$

$= n^2 - n + 1 + 2\sqrt{n(n-1)}$

Base $= \sqrt{n^2 - n + 1 + 2\sqrt{n}\sqrt{n-1}}$

$= \sqrt{n(n-1) + 1 + 2\sqrt{n}\sqrt{n-1}}$

$= \sqrt{\sqrt{n(n-1)^2} + (1)^2 + 2\sqrt{n}\sqrt{n-1}} = \sqrt{n(n-1)} + 1$

So, $\tan\theta = \frac{\sqrt{n}-\sqrt{n-1}}{1+\sqrt{n}\sqrt{n-1}} \Rightarrow \theta = \tan^{-1}\left(\frac{\sqrt{n}-\sqrt{n-1}}{1+\sqrt{n}\sqrt{n-1}}\right)$

$\theta = \tan^{-1}\sqrt{n} - \tan^{-1}\sqrt{n-1} = T_n$

$T_1 = \tan^{-1}\sqrt{1} - \tan^{-1} 0 \Rightarrow T_2 = \tan^{-1}\sqrt{2} - \tan^{-1}\sqrt{1}$

$T_3 = \tan^{-1}\sqrt{3} - \tan^{-1}\sqrt{2}$

$T_\infty = \tan^{-1}\sqrt{n} - \tan^{-1}\sqrt{n-1}$

Sum $= \tan^{-1}\sqrt{n} - \tan^{-1} 0 \Rightarrow n \to \infty$

Sum $= \tan^{-1}\sqrt{\infty} = \tan^{-1}\infty = \frac{\pi}{2}$

7. **(d)** $x^2 + y^2 + 2gx + 2fy = 0$

$C_1(-g, -f), r_1 = \sqrt{g^2 + f^2}$

$x^2 + y^2 + 2g'x + 2f'y = 0$

$C_2(-\square, -f'), r_2 = \sqrt{g'^2 + f'^2}$

When two circle touches, $r_1 + r_2 = C_1C_2$

$\sqrt{g^2+f^2} + \sqrt{g'^2+f'^2} = \sqrt{(g-g')^2 + (f-f')^2}$

Squaring both sides, we get

$g^2 + g'^2 + f^2 + f'^2 + 2\sqrt{g^2+f^2}\sqrt{g'^2+f'^2}$

$= g^2 + g'^2 - 2gg' + f^2 + f'^2 - 2ff'$

$(g^2+f^2)(g'^2+f'^2) = g^2g'^2 + f^2f'^2 + 2gg'ff'^2$

$g^2f'^2 + f^2g'^2 = 2gg'ff'$

$(gf' - fg')^2 = 0$

$gg' = ff'$

8. **(d)** $I = \int_0^\pi\ [\cot x]dx$

$I = \int_0^\pi\ [\cot(\pi - x)]dx$

$I = \int_0^\pi\ [-\cot x]dx$

Add Eqs. (i) and (ii), $2I = \int^\pi([\cot x] + [-\cot x])dx$

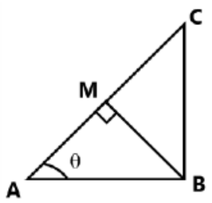
Using result $t[x] + [-x] = \begin{bmatrix} 0 & \text{if } x \in I \\ -1 & \text{if } x \notin I \end{bmatrix}$

$= \int_0^\pi\ -1dx$

2I $= -[x]_0^\pi \Rightarrow 2I = -\pi \Rightarrow I = -\frac{\pi}{2}$

9. **(c)** $MB = x$ (say)

$\Rightarrow\ AC = 4x \Rightarrow \cos\theta = \frac{AB}{4x}$



In $\triangle\ AMB$

$\Rightarrow \sin\theta = \frac{MB}{AB} = \frac{x}{AB} \Rightarrow \sin\theta \cdot \cos\theta = \frac{AB}{4x} \cdot \frac{x}{AB} = \frac{1}{4}$

$\Rightarrow 2\sin\theta\cos\theta = \frac{2}{4} = \frac{1}{2} \Rightarrow \sin 2\theta = \sin 30^\circ$

$\Rightarrow\ 2\theta = 30^\circ \Rightarrow \theta = 15^\circ$

10. (a) Probability of A being hit $= P(E) = 1 -$ (Probability of B missing) $\times$ (Probability of C missing)

$= 1 - \left(1 - \frac{1}{2}\right) \times \left(1 - \frac{1}{3}\right) = \frac{2}{3}$

Probability that A is hit by B and not $C = P[(B \cap \bar{C})/E]$

$= \frac{P(B) \times P(\bar{C})}{P(E)} = \frac{\frac{1}{2} \times \frac{2}{3}}{\frac{2}{3}} = \frac{1}{2}$

11. (a) Let $A = 2k, B = 3k, C = 7k$

In any triangle, $A + B + C = 180^\circ = \pi$

$12\text{k} = 180^\circ \Rightarrow k = 15^\circ$

A$= 30^\circ, B = 45^\circ, C = 105^\circ$

Using sine rule, $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

$\sin A : \sin B : \sin C = a : b : c$

$a : b : c = \sin 30^\circ : \sin 45^\circ : \sin 105^\circ$

$\left[\therefore \sin 105^\circ = \cos 15^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}}\right]$

$= \frac{1}{2} : \frac{1}{\sqrt{2}} : \frac{\sqrt{3}+1}{2\sqrt{2}} = \sqrt{2} : 2 : \sqrt{3} + 1$

12. (c) Conditional $\leq P(A \cap B) \leq \min\{P(A), P(B)\}$

Min $P(A \cap B) =$ when both are independent $P(A \cap B) = P(A)P(B) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$

Max $P(A \cap B) =$ when smaller set is a subset of larger set.

$P(A \cap B) = P(B) = \frac{1}{3} \Rightarrow$ Hence, $\frac{1}{6} \leq P(A \cap B) \leq \frac{1}{3}$

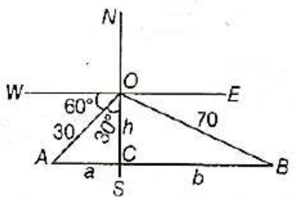
13. (d) One-to-one function

$n(A) = \{1,2,3\}$

$n(B) = \{1,2,3,4,5\}$

${}^5P_3 = = \frac{5!}{2!} = 5 \times 4 \times 3 = 60$

14. **(b)** $\sin 30^\circ = \frac{a}{30} \Rightarrow a = 30 \times \sin 30^\circ = 15$ km $\cos 30^\circ = \frac{OC}{30}$



$\Rightarrow OC = 15\sqrt{3}$

Now, using Pythagoras theorem in $\triangle OCB$

$h^2 + b^2 = 4900$

$\Rightarrow b = \sqrt{4900 - (15\sqrt{3})^2} = \sqrt{4900 - 675} = \sqrt{4225} = 65$

Now, $a + b = 65 + 15 = 80$

So, it will take $t = \frac{80}{10} =$ 8hrs and Noon $= 12:00$pm

So, after 8hrs, time will be 8: 00pm.

15. (d) $x = n - 1,\ n \in N,\ y = n$ and $z = n + 1$

$\log\left(1 + xz\right) = \log\left(1 + (n-1)(n+1)\right)$

$= \log\left(1 + (n^2 - 1)\right) = \log n^2 = 2\log n = 2\log y$

16. (c) If a, b, c are in geometric progression, then $b^2 = ac$ so by multiplying both sides by x^2,

$x^2b^2 = x^2ac \Rightarrow x^2b^2 = (xa)(xc)$

Taking log both sides of base x

$\log_x(x^2b^2) = \log_x[(xa)(xc)]$

$\Rightarrow 2\log_x(xb) = \log_x xa + \log_x xc$

Hence, $\log_x ax, \log_x bx, \log_x cx$ are in AP.

$\frac{1}{\log_{ax} x}, \frac{1}{\log_{bx} x}, \frac{1}{\log_{cx} x}$ are in AP

Hence, $\log_{ax} x, \log_{bx} x, \log_{cx} x$ are in HP.

17. (c) $-(2a+b).[(a-2b)\times(a\times b)]$

$= -(2a+b) - [\{(a-2b).b\}a - \{(a-2b).a\}b]$

$= (2a+b)[(a-2b).a]b - \{(a-2b) - b\}a]$

$= (2a+b)[\{|a|^2 - 2b.a\}b - \{a.b - 2|b|^2\}a]$

Now, $a.b = \left(\frac{\hat{i} - 2\hat{j}}{\sqrt{5}}\right).\left(\frac{2\hat{i} + \hat{j} + 3\hat{k}}{\sqrt{14}}\right) = 2 - 2 = 0$

$|a| = |b| = 1$

$2a + b.[\{1-0\}b - \{0-2\}a]$

$= (2a+b).[b+2a] = (2a+b).[b+2a]$

$= 2a.b + 4|a|^2 + |b|^2 + 2a.b = 0 + 4 + 1 + 0 = 5$

18. (b) $\frac{1}{2\sqrt{1}+1\sqrt{2}} + \frac{1}{3\sqrt{2}+2\sqrt{3}} + \frac{1}{4\sqrt{3}+3\sqrt{4}} + \cdots \ldots + \frac{1}{25\sqrt{24}+24\sqrt{25}}$

Its general term $T_n = \frac{1}{(n+1)\sqrt{n}+n\sqrt{n+1}}$

$= \frac{1}{(n+1)\sqrt{n}+n\sqrt{n+1}} \times \frac{(n+1)\sqrt{n}-n\sqrt{n+1}}{(n+1)\sqrt{n}-n\sqrt{n+1}}$

$= \frac{(n+1)\sqrt{n}-n\sqrt{n+1}}{n(n+1)^2-n^2(n+1)} = \frac{(n+1)\sqrt{n}-n\sqrt{n+1}}{n(n+1)}$

$T_n = = \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}$

Sum, $S = \sum_{n=1}^{24} T_n = \sum_{n=1}^{24}\left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}\right)$

$S = \left(\frac{1}{1} - \frac{1}{\sqrt{2}}\right) + \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{3}}\right) + \left(\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}}\right) + \cdots$

$+ \left(\frac{1}{\sqrt{24}} - \frac{1}{\sqrt{25}}\right)$

$S = \frac{1}{1} - \frac{1}{\sqrt{25}} \Rightarrow S = 1 - \frac{1}{5} \Rightarrow S = \frac{4}{5}$

19. (a) $\begin{vmatrix} 1 & 0 & -1 \\ x & 1 & 1-x \\ y & x & 1+x-y \end{vmatrix}$

$= \{(1+x-y)(x-x^2) - 0 - (x^2-y)\}$

$= 1 + x - y - x + x^2 - x^2 + y = 1$

Depends neither x nor y.

Other Method:

$\frac{dx}{dy} = \frac{1}{dy/dx} = \left(\frac{dy}{dx}\right)^{-1}$

$\frac{d}{dy}\left(\frac{dx}{dy}\right) = \frac{d}{dy}\left(\frac{dy}{dx}\right)^{-1} \times \left(\frac{dx}{dy}\right)$

$\frac{d^2x}{dy^2} = -\left(\frac{dy}{dx}\right)^{-2} \times \frac{d^2y}{dx^2} \times \left(\frac{dy}{dx}\right)^{-1}$

$\left[\frac{d^2x}{dy^2} = \left(\frac{dy}{dx}\right)^{-3} \times \left(\frac{d^2y}{dx^2}\right)\right]$

20. (c)

We have, $42\,{}^nP_2 = {}^nP_4$

$\Rightarrow 42\frac{n!}{(n-2)!} = \frac{n!}{(n-4)!} \Rightarrow \frac{(n-2)!}{(n-4)!} = 42$

$\Rightarrow \frac{(n-2)(n-3)(n-4)!}{(n-4)!} = 42 \Rightarrow n^2 - 5n + 6 = 42$

$\Rightarrow n^2-5n=36 \Rightarrow (n-9)(n+4)=0 \Rightarrow n=9$

Other method

Let u = $\log_{10}x$, v= $\log_x 10$

u. $\frac{\log x}{\log_{10}}$ v = $\frac{log_{10}}{log^x}$

$\frac{du}{dx}=\frac{1}{log_{10}}\frac{d}{dx}(\log x)$, $\frac{dv}{dx}=log_{10}\frac{d}{dx}\left(\frac{1}{log_x}\right)$

$\frac{du}{dx}=\frac{1}{log_{10}},\left(\frac{1}{x}\right)$, $\frac{dv}{dx}=log_{10}\frac{d}{dx}(\log_x)^{-1}$

21. (c) Let $\vec{a}$ and $\vec{b}$ two vector

(a) (a x b) and $\vec{a}$, it will be perpendicular to each other.

Some option (b) and (d) are perpendicular to each other.

Hence option (c) is correct.

22. (b) Because of A is a symmetric matrix.

So, $A^T=A$

$A\cdot A^T=I$

$$\begin{bmatrix}a&b&c\\b&c&a\\c&a&b\end{bmatrix}\begin{bmatrix}a&b&c\\b&c&a\\c&a&b\end{bmatrix}=I$$

$$\begin{bmatrix}a^2+b^2+c^2&ab+bc+ca&ac+ab+bc\\ab+bc+ac&b^2+c^2+a^2&bc+ca+ab\\ac+ab+bc&bc+ca+ab&c^2+a^2+b^2\end{bmatrix}=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$$

$a^2+b^2+c^2=1$

$ab+bc+ca=0$[$\therefore a,b,c$ are real and positive numbers.

23. (b) Equation of tangent with slope m

x= $a\cos 2t$

y = $2\sqrt{2}a\sin t$

$\frac{dx}{dt}=-2a\sin 2t$ and $\frac{dy}{dt}=2\sqrt{2}a\cos t$

$\frac{dy}{dx}=\frac{dy/dt}{dx/dt}=\frac{2\sqrt{2}a\cos t}{-2a\sin 2t}$

$=\frac{\sqrt{2}\cos t}{-2\sin t\cos t}=\frac{-1}{\sqrt{2}\sin t}=m$ (given)

$\sin t=\frac{-1}{\sqrt{2}m}$

Then,

x = $a\cos 2t$ y= $=2\sqrt{2}a\sin t$

$=a(1-2\sin^2 t)$ $=2\sqrt{2}a\times\left(\frac{-1}{\sqrt{2}m}\right)$

$=a\left(1-2\times\left(\frac{-1}{\sqrt{2}m}\right)^2\right)$ y = $=\frac{-2a}{m}$

x = $a\left(1-\frac{1}{m^2}\right)$

Then, equation of tangent

$y-y_1=m(x-x_1)$

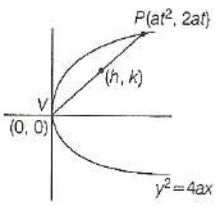
$y+\frac{2a}{m}=m\left(x-a\left(1-\frac{1}{m^2}\right)\right)$

$y+\frac{2a}{m}=mx-am+\frac{a}{m}$

y = $=mx-a\left(m+\frac{1}{m}\right)$

24. (b) Let (h, k) be the mid point of chord

S : $y^2=4x$



$h=\frac{(at^2+0)}{2},k=\frac{2at+0}{2}$

Hence, $t=\frac{k}{a}$

Put this value in $h=\frac{(at^2)}{2}$

$\Rightarrow 2h=\frac{ak^2}{a^2}$

$\Rightarrow 2ah=k^2$

Put $a=1$, we get $k^2=2h$

Hence, locus $y^2=2x$

25. (d) $\lim_{x\to a}\frac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3a+x}-2\sqrt{x}}$

$=\lim_{x\to a}\frac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3a+x}-2\sqrt{x}}\times\frac{\sqrt{a+2x}+\sqrt{3x}}{\sqrt{a+2x}+\sqrt{3x}}\times\frac{\sqrt{3a+x}+2\sqrt{x}}{\sqrt{3a+x}+2\sqrt{x}}$

$=\lim_{x\to a}\frac{\square+2x-3x}{3a+x-4x}\times\frac{\sqrt{3a+x}+2\sqrt{x}}{\sqrt{a+2x}+\sqrt{3x}}$

$=\lim_{x\to a}\frac{a-x}{3(a-x)}\times\frac{2\sqrt{a}+2\sqrt{a}}{\sqrt{3a}+\sqrt{3a}}\Rightarrow\lim_{x\to a}\frac{1}{3}\cdot\frac{4\sqrt{a}}{2\sqrt{3a}}=\frac{2}{3\sqrt{3}}$

26. (d) It is an even function

$\int_{-\pi/3}^{\pi/3}\frac{x\sin x}{\cos^2 x}dx=2\int_0^{\pi/3}\frac{x\sin x}{\cos^2 x}dx=2\int_0^{\pi/3}x\tan x\sec x dx$

Using integration by parts

$2\left([x\sec x]_0^{\pi/3}-\int_0^{\pi/3}1\times\sec x dx\right)$

We should know, $\int\sec x dx=\log|\sec x+\tan x|$

$=2\left[\left(\frac{\pi}{3}\sec\frac{\pi}{3}-0\right)-\log\left(\sec\frac{\pi}{3}+\tan\frac{\pi}{3}\right)\right.$

$+\log(\sec 0+\tan 0)]$

$=2\left(\frac{\pi}{3}\times 2\right)-2(\log(2+\sqrt{3}))$

We know that $\tan\frac{5\pi}{12}=2+\sqrt{3}=\frac{4\pi}{3}-2\log\tan\left(\frac{5\pi}{12}\right)$

27. (c)

$\frac{x^2}{16}+\frac{y^2}{b^2}=1$ (Ellipse); $\frac{x^2}{144}-\frac{y^2}{81}=\frac{1}{25}$ (Hyperbola)

a= 4, A= $\frac{12}{5},B=\frac{9}{5}$

$e^2=1-\frac{b^2}{a^2}$ $e^2=1+\frac{B^2}{A^2}$

$16-16e^2=b^2$

$e^2=1+\frac{81}{144}\Rightarrow\frac{15}{12}=e$

Focus of ellipse : $(ae,0)$ Focus of hyperbola : $(Ae,0)$

$(4e,0)$ $\left(\frac{12}{5}\times\frac{15}{12},0\right)=(3,0)$

Given : $(4e,0)=(3,0)$

$4e=3\Rightarrow e=\frac{3}{4}$

$\therefore b^2=16-16\left(\frac{9}{16}\right)$

$b^2=7$

28. (a). $\begin{vmatrix}\sin\pi&\sin B&\cos C\\-\sin B&0&\tan A\\\cos(\pi-C)&-\tan A&0\end{vmatrix}$

$=\begin{vmatrix}0&\sin B&\cos C\\-\sin B&0&\tan A\\-\cos C&-\tan A&0\end{vmatrix}$

Now, this determinant is skew symmetric of odd order Hence, its value will be zero

29. (d)

$1,1+d,1+2d,1+3d\ldots\ldots1+100d$

a= 1

Total number of terms $=101$

Sum $=\frac{n}{2}(a+l)=\frac{101}{2}(1+1+100d)=101(1+50d)$

Mean $=\frac{\sum x}{n}=\frac{101(1+50d)}{101}=1+50d$

Mean deviation $=\frac{\Sigma|\bar{x}-x_i|}{n}$

$=\frac{|1+50d-1|+|1+50d-1-d|+\cdots|1+100d-1-50d|}{101}$

$=\frac{50d+49d+48d+\cdots0+d+\cdots+50d}{101}=\frac{2d(1+2+\cdots+50)}{101}$

$\Rightarrow 225=\frac{50\times51\times d}{101}\Rightarrow d=\frac{225\times101}{50\times51}=\frac{9\times101}{102}=8.91$

30. (b)

$P = \sin^{20}\theta + \cos^{48}\theta$

$P = \sin^{20}\theta + \cos^{48}\theta > 0, \forall\theta$

Now, $P = \sin^{20}\theta + (1-\sin^2\theta)^{24}$

$0 \le \sin^2\theta \le 1$

$0 \le \sin^{20}\theta \le 1$

$\Rightarrow\ 0 \le \sin^{18}\theta \le 1$

$\Rightarrow\ 0 \le \sin^{18}\theta \le 1$

Now, $P = \sin^{20}\theta + (1-\sin^2\theta)^{24}$

Hence, $0 \le \sin^{20}\theta \le \sin^2\theta$

Similarly, $0 \le \cos^{46}\theta \cdot \cos^2\theta \le \cos^2\theta$

Add Eqs. (i) and (ii),

$\Rightarrow 0 < \sin^{20}\theta + \cos^{48}\theta \le \sin^2\theta + \cos^2\theta$

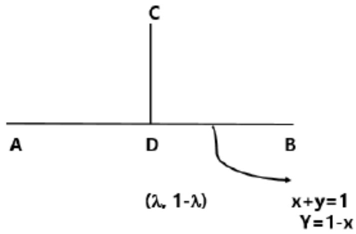
$\Rightarrow 0 < \sin^{20}\theta + \cos^{48}\theta \le 1$

$\Rightarrow 0 < P \le 1$

31. Let x coordinate of D is λ

y. l-λ

CD⊥AB (m₁m₂ =−1)



Slope of CD $=\frac{y_2-y_1}{x_2-x_1} = \frac{1-\lambda}{\lambda-2}$ $m_1 = -1$

= -3-λ

λ-2

$m_1 - m_2 = -1$

$-1 \times -\left(\frac{3+\lambda}{\lambda-2}\right) = -1$

3 + λ. 2−λ

2λ = −1

$x = \lambda = -\frac{1}{2}$

$y = 1+\frac{1}{2}$

$y = \frac{3}{2}$ (b) option

32. **(c)** Coefficient of x^2 cannot be equal to 0.

Hence, $(k-2) \ne 0 \Rightarrow k \ne 2$

Condition that both roots are real, distinct and (−) ve roots

$\Rightarrow D \ge 0$ and sum of roots < 0 and product of roots > 0.

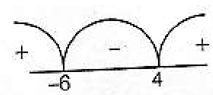
Sum of roots $= \alpha + \beta = \frac{-8}{k-2} < 0$

$\Rightarrow\ \frac{1}{k-2} > 0 \Rightarrow k > 2$

$D = (8)^2 - 4(k-2)(k+4) \ge 0$

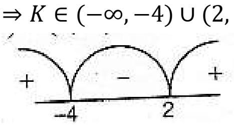
$\Rightarrow\ 16 - (k-2)(k+4) \ge 0 \Rightarrow k^2 + 2k - 24 \le 0$

$\Rightarrow\ (k+6)(k-4) \le 0 \Rightarrow k \in [-6,4]$



Product of roots $= \alpha\beta = \frac{k+4}{k-2} > 0$

$\Rightarrow K \in (-\infty, -4) \cup (2, \infty)$



Taking intersection of Eqs. (i), (ii), and (iii), we get

$2 < k \le 4$

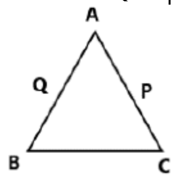
From given options the value of $k = 3$.

33. (2,1), (-1, -2) (3,3) are the mid points to sides

BC, CA, AB of ΔABC

Let the mid points of lie CA and AB are P and Q Slope of PQ

$m = \frac{y_0-y_1}{x_2-x_1} = \frac{3+2}{3+1} = \frac{5}{4}$



Line posing through (2, 1)

And parallel to line with slope 5/4.

$y - 1 = \frac{5}{4}(x-2)$

5x -4y – 6 = 0

34. **(d)**

Let a,b,c the event 1 appear on the die

So $P(A) = \frac{1}{6}, P(\bar{A}) = 1 - \frac{1}{b} = \frac{5}{6}$

Now, 1 appears in an even numbered throw

Required probability

$= P(\bar{A}). P(A) + P(\bar{A}). P(\bar{A}). P(\bar{A}). P(\bar{A})+.....$

$= \frac{5}{6} \times \frac{1}{6} + \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \cdots.$

$= \frac{5}{6} \times \frac{1}{6}\left[1 + \frac{25}{36} + \left(\frac{25}{36}\right)^2 \ldots.\right]$

$= \frac{5}{6}\left(\frac{1}{1-\frac{25}{36}}\right) \Rightarrow \frac{5}{11}$

35. A vector v in plane of a and b

so v = λa + μb

⇒ v= λ (i + j + k) + v (1-j+k)

= (λ+μ)i + (λ−μ)j + (λ+μ) R

Projection v and $\frac{c}{|c|}$ is $\frac{1}{\sqrt{3}}$

$\Rightarrow \frac{(\lambda+\mu)i+(\lambda-\mu)i+(\lambda+\mu)\hat{K},\frac{(i-j-\hat{K})}{\sqrt{3}}}{\left|\frac{c}{|c|}\right|} = \frac{1}{\sqrt{3}}$

$\frac{(\lambda+\mu)-(\lambda-\mu)-(\lambda-\mu)}{\sqrt{3}} = \frac{1}{\sqrt{3}}$

$\lambda - \mu = -1$

35. let $\bar{v} = a + \lambda\bar{b}$

$\bar{v}$= (I + j + $\hat{k}$)+ λ (i–j + $\hat{k}$)

(1+λ) I +(1-λ) j + (1+λ) $\hat{k}$

Projection of $\bar{v}$ on $\bar{c}$ = $\frac{V.C}{|C|} = \frac{1}{\sqrt{3}}$

$\frac{(1+\lambda)-(1-\lambda)(1+\lambda)}{\sqrt{3}} = \frac{1}{\sqrt{3}}$

λ– 1 = 1

λ = 2

$\bar{v}$=, 3i – j +3 $\hat{k}$

36 **(d)** Observe first 5 consecutive 0 's

Summation rule : the first 5 consecutive 0 's could start at position 1,2,3,4,5, or 6 .

Start at the first position.

Other 5 bits can be anything $2^5 = 32$

Start at the second position.

The first bit must be a 1 .

There are various possibilities that can be included in it.

Remaining bits can be anything $2^4 = 16$

Start at third position.

The second bit must be a 1 due to above-mentioned reason.

First bit and last 3 bits can be anything $2^4 = 16$

Starting at 4,5 and 6 positions.

Same as starting at positions 2 or 3,16 each

Total $= 32 + 16 + 16 + 16 + 16 + 16 = 112$

The five consecutive 1's ensue the same pattern and have different like 112 possibilities.

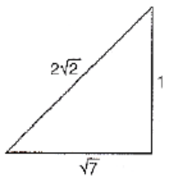
There would be two cases counted twice (that we thus need to excluded) 0000011111 and 1111100000 .

Total $= 112 + 112 - 2 = 222$

37. (b)

$\cos x + \sin x = \frac{1}{2}[0 < x < \pi]$

$\Rightarrow \sqrt{2}\left[\frac{1}{2}\cos x + \frac{1}{\sqrt{2}} sin x\right] = \frac{1}{2} \Rightarrow sin\left(x + \frac{\pi}{4}\right) = \frac{1}{2\sqrt{2}}$



$\Rightarrow \tan\left(x + \frac{\pi}{4}\right) = \frac{1}{\sqrt{7}} \Rightarrow \frac{\tan x + \tan\frac{\pi}{4}}{1 - \tan x \tan\frac{\pi}{4}} = \frac{\tan x + 1}{1 - \tan x} = \frac{1}{\sqrt{7}}$

$\Rightarrow \sqrt{7}\tan x + \sqrt{7} = 1 - \tan x$

$\Rightarrow (1 + \sqrt{7})\tan x = 1 - \sqrt{7}$

$\Rightarrow \tan x = \frac{1-\sqrt{7}}{1+\sqrt{7}} = \frac{(1-\sqrt{7})^2}{1-7}$

$= \frac{1+7-2\sqrt{7}}{-6} = \frac{2(4-\sqrt{7})}{-6} = \frac{\sqrt{7}-4}{3}$

38. (*a*) We have *a*, *b* and *c* are the position vectors of the vertices of A, B and C of a ΔABC.

$= \frac{1}{2}(b - a) \times (c \times a) \qquad \frac{1}{2}|a \times b + b \times c + c \times a|$

39. **(c)** $\int e^x(f(x) - f'(x))dx = \phi(x)$

$\int e^x f(x)dx = ?$

$\int e^x f(x)dx - \int e^x f'(x)dx = \phi(x)$

Using integration by parts,

$\int e^x f(x)dx - [e^x \int f'(x)dx - \int e^x \int f'(x)dx] = \phi(x)$

$\int e^x f(x)dx - e^x f(x) + \int e^x f(x)dx = \phi(x)$

$2\int e^x f(x)dx = \phi(x) + e^x f(x)$

$\int e^x f(x)dx = \frac{1}{2}[\phi(x) + e^x f(x)]$

40. **(a)** $3x + 4y + K = 0$ is a tangent to the hyperbola

$9x^2 - 16y^2 = 144$

$\frac{x^2}{16} - \frac{y^2}{9} = 1$

$a = 4, b = 3$

Given tangent $y = \frac{-3}{4}x - \frac{K}{4}$

Let the line be, $y = mx + c$

$\therefore\ m = \frac{-3}{4},\ c = \frac{-K}{4}$

Condition of Tangency: $c^2 = a^2m^2 - b^2$

$\frac{K^2}{16} = 16 \times \frac{9}{16} - 9$

$\therefore$ K = 0

41. (c)

$a^2 + 2b^2 - 2bc = 100$

And

$2ab - c^2 = 100$

On subtracting Eq. (ii) from Eq. (i), we get

$a^2 + 2b^2 - 2bc - 2ab + c^2 = 100 - 100$

$\Rightarrow a^2 + b^2 - 2ab + b^2 + c^2 - 2bc = 0$

$\Rightarrow (a - b)^2 + (b - c)^2 = 0 \Rightarrow a - b = 0$ and $b - c = 0$

$\Rightarrow a = b$ and $b = c \Rightarrow a = b = c = k$ (say)

$\Rightarrow \frac{a+b}{c} = \frac{2k}{k} = 2$

42. The equation of perpendicular line to 7x – y + 8 = 0 is x + 7 y = λ ....(1)

–4 + 7 (5) = λ

λ = 31

Equation will be, x + 7 y = 31

43. **(d)** There are total $2n + 1$ tickets in which n tickets are even numbered and $n + 1$ tickets are odd numbered. Now, we have to select three tickets which are in AP.

Let a, b, c are numbers written on the 3 tickets which are drawn.

If these are in AP then $2b = a + c$ means $b = (a + c)/2$ a, b, c are natural numbers Hence b is also a natural number. Means $a + c$ must be divisible by 2 .

Hence, probability that 3 tickets drawn are in

AP $= \frac{{}^nC_2 + {}^{n+1}C_2}{{}^{2n+1}C_3}$

$= \frac{3n}{4n^2 - 1}$

44. Let C_1 (h, k) be the center of the circle touches the x axis then into radius is r_1 = k

Circle C_1 touches another circle C_2 (0,2) with radius r_2 = 2

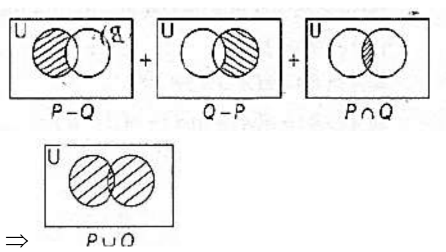
$|C_1C_2| \Rightarrow r_1 + C_2$

$\sqrt{(h - o)^2 + (K - 3)^2}$ = K + 2

h^2– 10K + 5 = 0, replacing h, k with a, y

Locus is x^2– 10y + 5 = 0

45. **(a)**



46. (d)

$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \left(\frac{dy}{dx}\right)^{-1}\frac{dy}{dx}$

$\frac{d}{dy}\left(\frac{dx}{dy}\right) = \frac{d}{dx}\left(\frac{dy}{dx}\right)^{-1}\frac{dy}{dx}$

$\frac{d^2x}{dy^2} = \left(\frac{d^2y}{dx^2}\right)\text{-}\left(\frac{dy}{dx}\right)^{-2}\frac{dy}{dx}$

$\frac{d^2x}{dy^2} = \left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$

47. **(a)** Ellipse : $\frac{x^2}{16} + \frac{y^2}{9} = 1$

$a^2 = 16, b^2 = 9 \Rightarrow b^2 = a^2(1 - e^2) \Rightarrow 9 = 16(1 - e^2)$

$e^2 = 1 - \frac{9}{16} \Rightarrow e = \frac{\sqrt{7}}{4}$

Foci $(\pm\sqrt{7}, 0)$

$\therefore$ Circle passes through $(\sqrt{7}, 0)$& having centre $(0,3)$

$\therefore r = \sqrt{(\sqrt{7})^2 + (3)^2} = \sqrt{7+9} = 4$ units

48. **(c)** A function $F: (0, \pi) \to R$ defined

by $f(x) = 2\sin x + \cos 2x$ has

$f'(x) = 2\cos x - 2\sin 2x = 0$

$\Rightarrow 2\cos x - (2\cos x \sin x)2 = 0$

$\Rightarrow 2\cos x(1 - 2\sin x) = 0 \Rightarrow \cos x = 0$

$\Rightarrow x = \frac{\pi}{2}$

And $\sin x = \frac{1}{2}$

$\Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$

$f''(x) = -2\sin x - 4\cos 2x$

$f''\left(\frac{\pi}{2}\right) = = -2 \times 1 - 4\cos 2 \times \frac{\pi}{2}$

$= -2 - 4 \times (-1) = 2 \to$ minima

$f''\left(\frac{\pi}{6}\right) = -2 \times \frac{1}{2} - 4\cos 2 \times \frac{\pi}{6}$

$= -1 - 4 \times \frac{1}{2} = -3 \to$ maxima

$f(x)$ has both local minima and local maxima.

49. $M_r = \begin{bmatrix} r & r-1 \\ r-1 & r \end{bmatrix} r \in N$

$|M_r| = \begin{bmatrix} r & r-1 \\ r-1 & r \end{bmatrix} \quad r \in N$

$|m_r| = 2r - 1$

det (M_1) + det (M_2) + ... + det (M_{2015})

$\sum_{r=1}^{2015} |Mr| = 2\sum_{r=1}^{2015} \quad r - 2015$

$\sum_{r=1}^{2015} |Mr| = \frac{2 \times 2015 \times 2016}{2} - 2015$

$= (2015)^2$

50. **(a)** Area of quadrilateral $= \frac{|AC \times BD|}{2}$

$AC \times BD = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 1 \\ -1 & 3 & 2 \end{vmatrix} = \hat{i}(-1) - \hat{j}(4+1) + \hat{k}(6+1)$

$= -\hat{i} - 5\hat{j} + 7\hat{k}$

Area $= \frac{|-\hat{i} - 5\hat{j} + 7\hat{k}|}{2}$

$= \frac{5\sqrt{3}}{2}$

REASONING

Solution [Q. Nos. 51 to 54]

The given information can be represented as:

Stage	Farmer	Starting point of the stage	Ending point of the stage
First	F_3	P_2	P_5
Second	F_1	P_5	P_3
Third	F_2	P_3	P_1
Fourth	F_4	P_1	P_4
Fifth	F_5	P_4	P_2

51. (a) P is the finishing point of

52. (d) 5th stage is ploughed by

53. (b) P and P

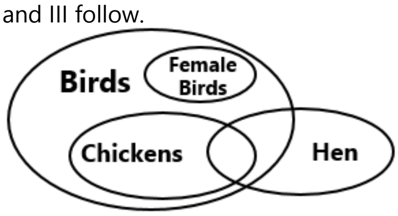
54. (b) P is the starting point of stage 3.

55. (c) The 4th day is Tuesday, then 11th, 12th and 25th also Tuesdays.

Then, 3 days after 24th is 27th, which is Thursday.

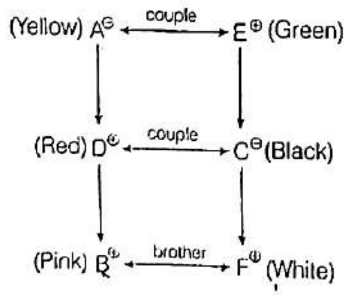
56. (b)

So, only conclusion II and III follow.



Solutions [Q. Nos. 57 to 60]

From the given information, we can draw a relation diagram,



57. (b) A likes yellow colour.

58. (d) Yellow-Hypen green is the colour combination of AE

59. (a) CD is the married couple.

60. (a) F is the brother of B.

61. (a) As,

E	X	A	M	I	N	A	T	I	O	N
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
5	6	1	4	9	5	1	2	9	6	5

The pattern in the above word is as follows.

E = 5

X = 24 = 2 + 4 = 6

A = 1

M = 13 = 1 + 3 = 4

1 = 9

N = 14 = 1 + 4 = 5

A = 1

T = 20 = 2 + 0 = 2

1 = 9

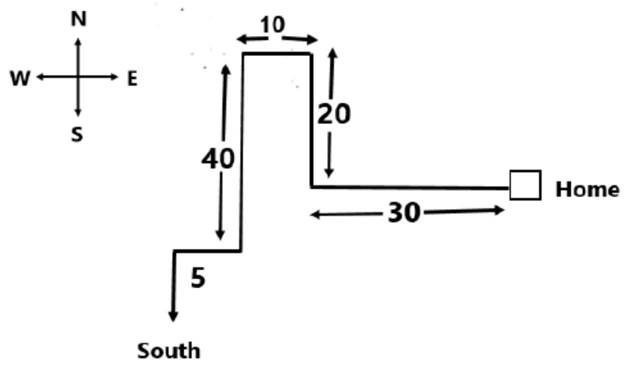
O = 15 = 1 + 5 = 6

N = 14 = 1 + 4 = 5

$\therefore$ The sum of the placing value of the words will be the code of the words. Similarly,

G	O	V	E	R	N	M	E	N	T
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
7	6	4	5	9	5	4	5	5	2

62. (b) According to the information given in the question, we can draw a direction diagram.



From the above diagram, it is clear that Gopal is walking in South direction.

63. (d)

So, no conclusion follows.

64. (d) 8, 6, 9, 23, 87 ?
The pattern is as follows
8 × 1 – 2 = 6
6 × 2 – 3 = 9
9 × 3 – 4 = 23
23 × 4 – 5 = 87
87 × 5 – 6 = 429
So, the next term will be 429.

65. (a) Suppose, the number of questions in group A, B and C be x, y, and z, respectively.
Then, x + y + z = 100 (given)
Total marks would be x + 2y + 3z
Given, y = 23
Total marks from section B = 46
Different possible values for z are 1, 2, 3......
∴ Corresponding values for x are 76, 75, 74...
(Since, total questions are 100)
when z = 1, x = 76, y = 23
Total marks from groups A, B and C are 76, 46, 3 respectively.
Percentage marks from A = $\frac{75}{76+46+3}$
$= \frac{75}{125} < 60\%$
when. z = 2, x = 75, y = 23
Percentage marks from A = = $\frac{75}{76+46+6}$
$= \frac{75}{127} < 60\%$
For all other values, when z increases, x decreases and contribution of marks from A keeps decreasing.
∴ There is only 1 possible value for questions from group C.

66. (a) Given,

÷	→	+
–	→	÷
×	→	–
+	→	×

Then,

$$\frac{(36 \times 4) - 8 \times 4}{4 + 8 \times 2 + 16 \div 1}$$

$$= \frac{(36 - 4) \div 8 - 4}{4 \times 8 - 2 \times 16 + 1}$$

$$= \frac{(32) \div 8 - 4}{4 \times 8 - 2 \times 16 + 1}$$

$$= \frac{4 - 4}{4 \times 8 - 2 \times 16 + 1} = 0$$

67. (c) In the word CYBERNETICS,
I occupies the same position as it does in the English alphabet.
The place value of 'I' is 9.

68. (c) Here, 2^{31} is divided by 5
$2^{31} = 2^{30} \times 2^1$
$= (2^2)^{15} \times 2$
$= 4^{15} \times 2$
$= (-1 + 5) \times 2 = 4 \times 2 = 8 \div 5 = 3$

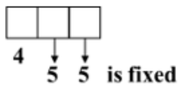
Solutions [Q. Nos. 19 to 20]

AN BO CP D Q E R F S G T HU IV JW KXLY MZ

69. (c) 7th letter from the left is D and 3rd letter to the right of D would be 'R'.

70. (b) The eighteenth letter from the beginning is V and fifteenth letter from the last is S. So, H will be exactly in between S and V.

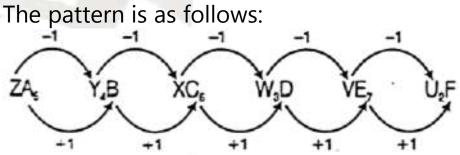
71. (b) Here



So, total 3 digits number divisible by 5
= 5 × 4 × 1 = 20

72. (a) Here,
We have, to find LCM of 78 paise, 69 paise and 101 paise.
And,
78 = 1 × 50 + 2 × 10 + 2 × 4 = 5 coins
69 = 1 × 50 + 1 × 10 + 1 × 5 + 2 × 2 = 5
101 = 1 ×101 = 1 × 50 + 4 × 10 + 1 × 5 + 3 × 2 = 9
So, LCM = 2 × 3 × 13 × 23 × 101 =181194

73. (c) ZA_5, Y_4B, XC_6, W_3D - -
The pattern is as follows:



So, VE, and U_2F will be fit in the blanks.

74. (d) 61, 52, 63, 94?
The number formed by reversing the digit of each number of the sequence is square of consecutive natural numbers starting from 4. See how,
61 .... 16 = 4^2
52 .... 25 = 5^2
63 .... 36 = 6^2
94 .... 49 = 7^2
Therefore, the required number = 8^2 = 64 = 46.

75. (d) From the given information

Wife	Husband
X	A (Lawyer)
Y	B (Doctor)
Z	C (Engineer)

From the above table, no statement is correct.

76. (d) From the statements,
A is placed below E

E
A

C is placed below D

D
C

B is placed below A

A
B

D is placed between A and E, then

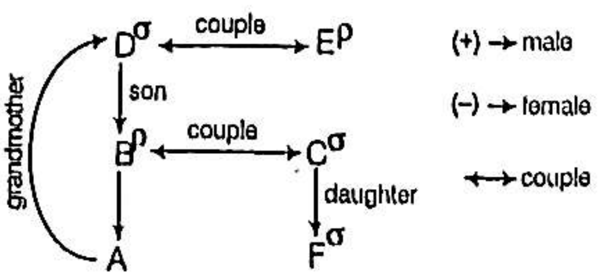
E
D
A
B
C

So, E will be on the top.

77. (c) Given that
B is taller than E but shorter than D.
D > B > E
A is shorter than C but taller than D.
D < A < C
Then the order is
C > A > D > B > E
1 2 3 4 5
So, B is fourth.

Solutions [Q. Nos. 78 to 81]

From the given information in the question, we can draw a relation diagram,



78. (c) 'C' is the mother of A

79. (d) It cannot be determined because the gender of A is not defined.

80. (d) DF is the couple.

81. (d) No statement is true.

Directions [Q. Nos. 82-85]

The given information in the question can be summarised as:

Girl	Money	Item
A	10	Lollipops
B	200	Toffees
C	60	Chocolates
D	40	Lollipops
E	120	Chocolates
F	80	Toffees
G	20	Chocolates

82. (a) From the table, it is clear that G has Rs.20.

83. (b) From the table, it is clear that C, G and E has chocolates with them.

84. (a) Combination in option (a) is correct.

85. (c) From the table, it is clear that D has Rs.40 with her.

86. (a) From the statement I
1-T and U are sitting at extreme ends of the row.
Case I T ..... U
Case II U I
From statement II
S is third to the right of T.
Then, from case I
T ...... S ...... U
So, S is sitting exactly in the middle.
So, only statement I and II are sufficient.

87. (b) The hands of a clock coincide 11 times in every 12 h (Since, between 11 and 1, they coincide only once i.e. at 12 O'clock).

AM	PM
12:00	12:00
1:05	1:05
2:11	2:11
3:16	3:16
4:22	4:22
5:27	5:27
6:33	6:33
7:38	7:38
8:44	8:44
9:49	9:49
10:55	10:55

The hands overlap about 65 min, not every 60 min. So, the hand coincide 22 times in a day.

88. (a) As,
As,

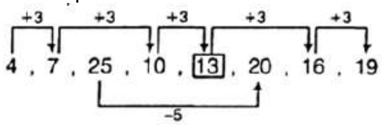
T	O	G	E	T	H	E	R
–2	+2	–2	+2	–2	+2	–2	+2
R	Q	E	G	R	J	C	T

Similarly,

P	A	R	O	L	E
N	C	P	Q	J	G

89. (c) Even if first 2 are different colours, 3rd one has to be black or brown. So, the smallest number of socks will be 3.

90. (a) 4, 7, 25, 10 _ 20, 16, 19
The pattern is as follows:



Two series alternate here, with every third number following a different patterns. In the main series, 3 is added to each number to arrive at the next. In the alternating series, 5 is subtracted from each number to arrive at the next.

ENGLISH

91. A	92. b	93. b	94. a	95. b	96. c	97. a
98. D	99. b	100. b	101.b	102. a	103. b	104. a
105. C	106. b	107. c	108. b	109. a	110. d	

COMPUTER

111. **(d)** 37H means $(00110111)_2$
17H means $(00010111)_2$
Now, we convert 37H, 17H binary number into decimal number.
$(00110111)_2$ decimal number $= (55)_{10}$
$(00010111)_2$ decimal number $= (23)_{10}$
$(55)_{h_0} + (23)_{h_0} = 23)55(2 \quad = (9)_{10}$
$\underline{46}$ or
$\underline{9}$ $(1001)_2$
$(1001)_2$ Hexadecimal $= 09$, remainder $= 09$H On division of 37H by 17H the remainder is 09H.

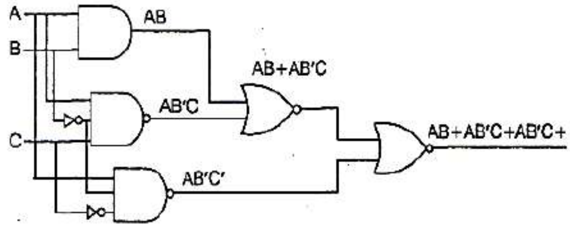
112. **(b)** If there are p possibilities for choice 1 and q possibilities for choice 2, then there are a total of p^*q different way of doing both. It is trivial that this can be extended to n choices. So, there would be 2^n boolean functions.

113. **(b)**

Binary	h	$h_3h_2h_1h_0$	$(n+1)$ mod16	$g_3g_2g_1g_0$
0000	0	0000	1	0001
0001	1	0001	2	0011
0010	2	0011	3	0010
0011	3	0010	4	0110
0100	4	0110	5	0111
0101	5	0111	6	0101
0110	6	0101	7	0100
0111	7	0100	8	1100
1000	8	1100	9	1101
1001	9	1101	10	1111
1010	10	1111	11	1110
1011	11	1110	12	1010
1100	12	1010	13	1011
1101	13	1011	14	1001
1110	14	1001	15	1000
1111	15	1000	0	0000

This gives the solution option (b).
$g_1(h_3h_2h_1h_0) = \Sigma(4,9,10,11,12,13,14,15)$

114. **(b)**



As, we can see in Logic Gates.
There are two NAND required.

115. **(d)** The diffraction beam an be machine read by a reading device which produces a reading light beam directed at the grating. The reading device has a plurality of detectors positioned to be illuminated by the diffracted light beam information can be stored in the optical memory by locally changing the optional properties of the grating.

116. **(a)**

(a) P= $(F87B)_{16}$ = 1111100001111011
2's complement = 1111100001111011
+1
0000011110000101
$(0000 \; 1 \; 1 \; 1 \; 1 \; 0 \; 0 \; 0 \; 0 \; 1 \; 0 \; 1)_2$
↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓
1024 512 256 128 64 32 16 8 4 2 1

Value in decimal is $(1925)_{10}$, Hence, $P = -1925$
So, now doing $8^*P = (-15400)_{10}$
Since, a negative number we have to find the 2 's complement of -15400
Binary of 15400 = 0011110000101000
2's complement = 1100 0011 1101 1000
C 3 D 8
= $(C3\ D8)_{16}$

117. **(c)**

From the diagram
$f \quad = f_1f_2 + f_3$ (put value)
$= (4,5,6,7,8)f_2 + (1,6,15) = (1,6,8,15)$
Now, as you know that the number 0,1,2, ... 14,15 represent the minterms like $\Sigma_m(6,8)$.
(i) If you AND (4,6) with (4,5,6,7,8) it will result in (4,6) but we don't need 4, so option (a) is wrong.
(ii) If you AND (4,8) with (4,5,6,7,8) it will result in (4,8) but we don't need 4, so option (b) is wrong.
(iii) If you AND (4,6,8) with (4,5,6,7,8) it will result in (4,6,8) but we don't need 4 here too, so option (d) is wrong.
(iv) If you AND (6,8) with (4,5,6,7,8) it will result in (6,8) now when $OR(6,8)$ with (1,6,15), you get (1,6,8,15) which is f, so option (c) is correct

118. **(d)**

Given, expression $(\overline{\overline{x+y}+\bar{z}})$

As, we know De-morgan's law says

$\overline{a+b}=\bar{a}\cdot\bar{b}$

So, let

$x+y=a$

$z=b$

$\Rightarrow \overline{(\overline{x+y}+\bar{z})}=\overline{\overline{x+y}}\cdot\bar{\bar{z}}$

$\frac{\downarrow\downarrow}{a+b}\ \frac{\downarrow}{a}\cdot\frac{\downarrow}{b}$

$\Rightarrow \overline{\overline{x+y}}\cdot\bar{\bar{z}}=(x+y)\cdot z$

Hence, $\overline{\overline{x+y}+\bar{z}}=(x+y)\cdot z$

119. **(c)** Since, $q\rightarrow s$ and $r\rightarrow s$, thus $(q\vee r)\rightarrow s, p\rightarrow(q\vee r)$ is given, then by rule of 'Syllogism'. $p\rightarrow s$ is deducted.

120. **(a)** Dynamic random access memories (DRAMs) are semiconductor integrated circuits (ICs) that operate like a bank of capacitors. DRAM consist atleast 1 MOS transistor necessarily.

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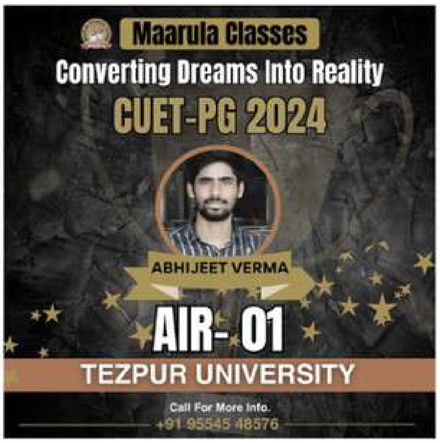

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